

Highlights

- AW-DPCNN fuses STFT and GADF images via contrast-guided dynamic weighting and dual-channel stimuli
- MSCA-VGG16 with multi-scale convolution, channel attention, and compact embedding for diagnosis
- Achieves 99.85% accuracy on CWRU 12 kHz drive-end bearing fault benchmark
- Demonstrates cross-sensor and cross-sampling-rate generalization with systematic ablation
- Maintains 91.2% accuracy at 5.0 dB SNR under additive Gaussian noise